

Parameter θ + resolved experiment
Response, exposure, noise and selection are specified

Joint measurement and reporting law
Generate Y and the reporting event D

$D = 0$: successful nondetection
Record the flag and response
Likelihood: $P(D = 0 \mid \theta)$

$D = 1$: retained measurement y
Record values and covariance semantics
Likelihood: $p(y, D = 1 \mid \theta)$

Use exactly the acquired record
Do not add unreported astrometry or count selection twice
Forced photometry requires a different record contract

Information anchor
If every θ always produces the same retained record,
 $I(\theta; \text{record}) = 0$